DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech. **Branch: Common to all branches** **Semester: I**

Subject Code & Name: (BTBS 101) Engineering Mathematics - I

Max Marks: 60 **Date: 06/02/2025** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q.1	**Objective type questions. (Compulsory Question)**		**12**
1	The eigen values of the matrix $A = \begin{bmatrix} a & h & g \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$ are a. a, b, g b. a, b, 0 c. a, 0, 0 d. a, b, c	Remember (CO1)	1
2	If λ is an eigen value of the matrix A, then the eigen value of A^{-1} is a. $\frac{1}{\lambda}$ b. λ c. $\lambda - 1$ d. None	Remember (CO1)	1
3	If $A = [a_{ij}]$ is a square matrix of order n, then trace of matrix A is a. product of diagonal elements b. sum of diagonal elements c. sum of row elements d. None	Remember (CO1)	1
4	If $u = x^y$ then $\frac{\partial u}{\partial x}$ is equal to a. 0 b. yx^{y-1} c. $x^y \log x$ d. None	Remember (CO2)	1
5	If $u = \log(x^2 + y^2 + 2xy)$ then a. $u_{xy} = u_{yx}$ b. $u_{xx} = u_{yx}$ c. $u_{xy} = u_{yy}$ d. None	Remember (CO2)	1
6	If J and J' are Jacobian and its inverse Jacobian respectively then which of the following statement is true? a. $J = J'$ b. $J * J' = 0$ c. $J * J' = 1$ d. None	Remember (CO3)	1
7	The condition for the function $f(x, y)$ to have maximum value at (a, b) is a. $rt - s^2 > 0 \, and \, r < 0$ b. $rt - s^2 > 0 \, and \, r > 0$ c. $rt - s^2 < 0 \, and \, r < 0$ d. None	Remember (CO3)	1
8	If u and v are continuous functions and differentiable functions of two independent variables x and y then the Jacobian of u, v w. r. t. x and y is a. $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial y} & \frac{\partial u}{\partial y} \end{vmatrix}$ b. $\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial y}{\partial v} & \frac{\partial x}{\partial v} \end{vmatrix}$ c. $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ d. $\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$	Remember (CO3)	1

9	The curve $xy = c$ is symmetrical about a. X-axis | b. Y-axis | c. $y = x$ | d. None	Remember (CO4)	1
10	The cartesian curve is symmetric about x- axis if a. Powers of x are even | b. Powers of y are even | c. Powers of x are odd | d. None	Remember (CO4)	1
11	The value of integral $\int_0^{\frac{\pi}{2}}\int_0^2 r dr d\theta$ is a. $\frac{\pi}{2}$ | b. π | c.1 | d. None	Remember (CO5)	1
12	In polar co-ordinate system (r,θ) ; value of $dy\ dx =$ a. $dr\ d\theta$ | b. $rdr\ d\theta$ | c. $r^2 dr\ d\theta$ | d. None	Remember (CO5)	1
Q.2	**Solve the following.**		12
A)	Find the rank of matrix by converting it into normal form $A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$	Understand (CO1)	6
B)	Find eigen values and eigen vector for greatest eigen value $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$	Understand (CO1)	6
Q.3	**Solve the following.**		12
A)	If $u = \sin^{-1}(x^2 + y^2)$ then find the value of $x^2 u_{xx} + 2xyu_{xy} + y^2 u_{yy}$	Remember (CO2)	6
B)	If $u = (1 - 2xy + y^2)^{\frac{-1}{2}}$ then show that $x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} = y^2 u^3$	Understand (CO2)	6
Q.4	**Solve any TWO of the following.**		12
A)	Find the Jacobian of $u = x^2 - 2y, v = x + y + z, w = x - 2y + 3$	Understand (CO3)	6
B)	Find the expansion of function $f(x,y) = e^{xy} at\ (1,1)$	Understand (CO3)	6
C)	Find three parts by dividing 120 into three parts so that the sum of their products taken two at a time shall be maximum.	Understand (CO3)	6
Q.5	**Solve any TWO of the following.**		12
A)	Solve $\int_0^2 x^3\sqrt{(2-x)}dx$ $\int_0^2 x^3\sqrt{(2-x)}dx$	Apply (CO4)	6
B)	Identify and trace the curve $x = a(\theta - sin\theta), y = a(1 + cos\theta)$	Apply (CO4)	6
C)	Identify and trace the curve $r = a(1 + cos\theta)$	Apply (CO4)	6
Q.6	**Solve any TWO of the following.**		12
A)	Solve $\int_0^1\int_0^y\int_0^{y+z} dx\ dy\ dz$	Apply (CO5)	6
B)	Find the area which lies outside the circle $r = 2$ and inside the cardioid $r = 2(1 + cos\theta)$	Remember (CO5)	6
C)	Define and verify Cayley Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -1 \end{bmatrix}$	Remember (CO2)	6
	***** End *****		